EXPERIMNET-1

Data Acquisition and Data Profiling using BI Tools

Aim: To acquire data from multiple sources and perform data profiling using Microsoft Power BI in order to assess data quality, understand data structure, identify anomalies, and prepare data for further analytical processing.

Tools / Software Required:

- Microsoft Power BI Desktop
- Sample datasets: Sales_Data.csv, Employee_DB.xlsx, and a sample SQL database

Theory:

Business Intelligence (BI)

Business Intelligence refers to a set of technologies, processes, and practices that enable organizations to collect, integrate, analyse, and present business information. BI helps transform raw data into meaningful, actionable insights that support better decision-making across an organization.

Key components of a BI system include:

- Data Sources: Databases (SQL/NoSQL), flat files, cloud stores, and APIs
- ETL Pipeline: Extract, Transform, Load processes that move data from source to destination
- Data Warehouse / Data Lake: Centralized repositories for structured or unstructured data
- BI & Reporting Tool: Microsoft Power BI for data connection, transformation, and visualization
- Analytics Layer: Descriptive, diagnostic, predictive, and prescriptive analytics

Data Acquisition

Data Acquisition is the process of collecting and importing data from various heterogeneous sources into a unified environment for processing. It is the first and foundational step in any BI pipeline. In Power BI, data is acquired through the Get Data option, which supports a wide range of connectors including flat files (CSV, XLSX), relational databases (SQL Server, MySQL), cloud sources (SharePoint, Azure), and web-based sources.

Data Profiling

Data Profiling is the process of examining data from an existing source and collecting statistics, summaries, and metadata about that data. It is a critical quality assurance step that reveals the structure, content, and quality of a dataset before further processing. Data profiling covers the following key dimensions:

- Completeness: Percentage of non-null values in each column.
- Uniqueness: Ratio of distinct to total values — detects duplicate records.
- Validity: Whether values conform to defined rules or expected formats.

- Consistency: Uniformity of values across related datasets.
- Accuracy: Correctness of values relative to real-world facts.
- Timeliness: Currency of data relative to the reporting period.

Data Profiling in Power BI — Power Query Editor

Microsoft Power BI provides built-in data profiling capabilities through the Power Query Editor. Under the View tab, three profiling features can be enabled:

- Column Quality: Displays the percentage of valid, error, and empty values per column.
- Column Distribution: Shows a histogram of value frequency along with the distinct and unique count for each column.
- Column Profile: Provides detailed statistics for a selected column including minimum, maximum, mean, median, standard deviation, and a full value distribution chart.

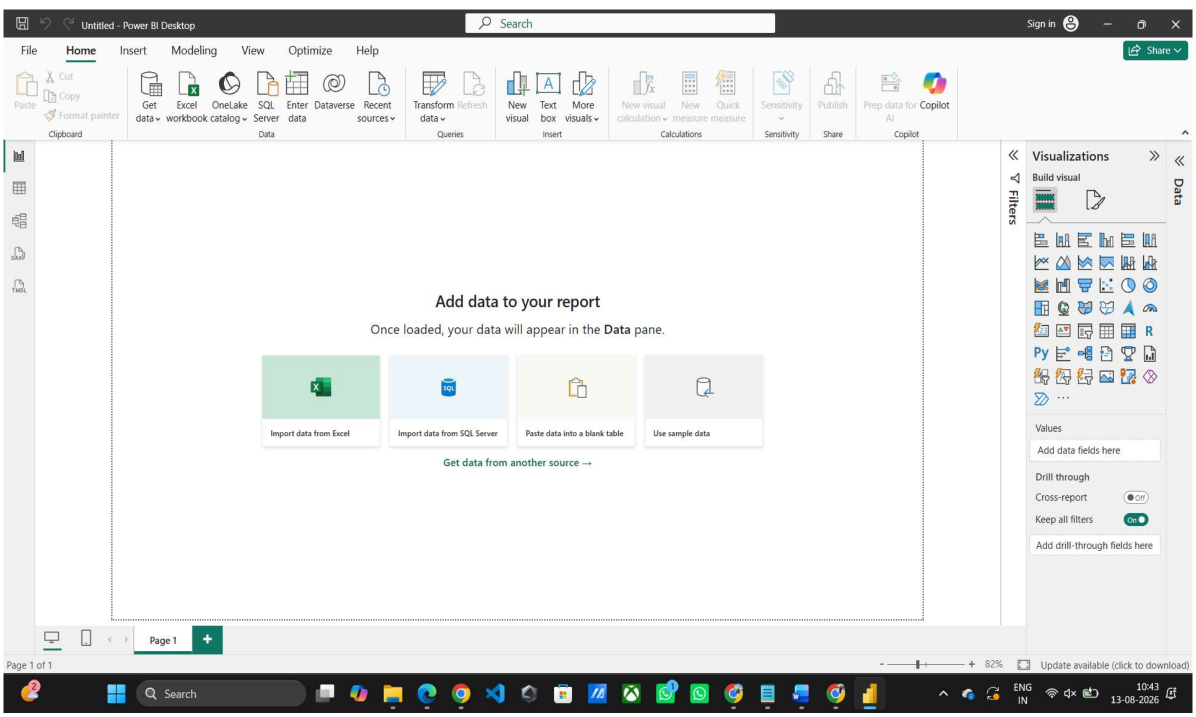
Procedure :

Part A: Software Setup and Data Acquisition

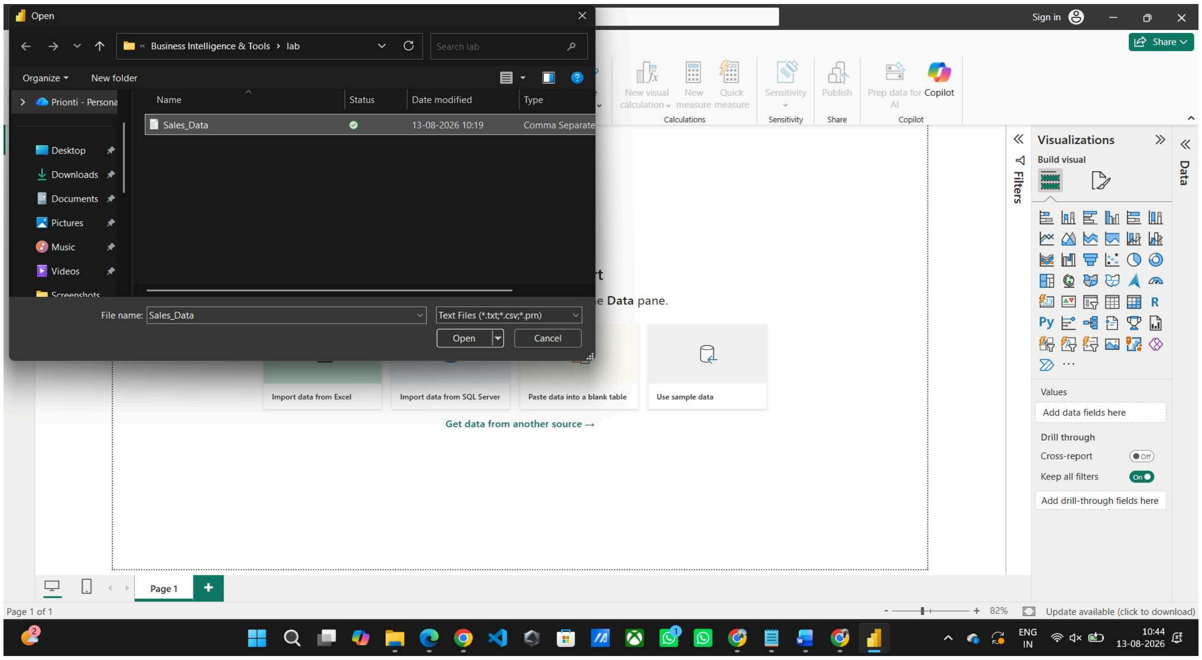
Step 1: Install the required software.

- Download and install Microsoft Power BI Desktop.

Step 2: Launch Power BI Desktop and import the CSV file.

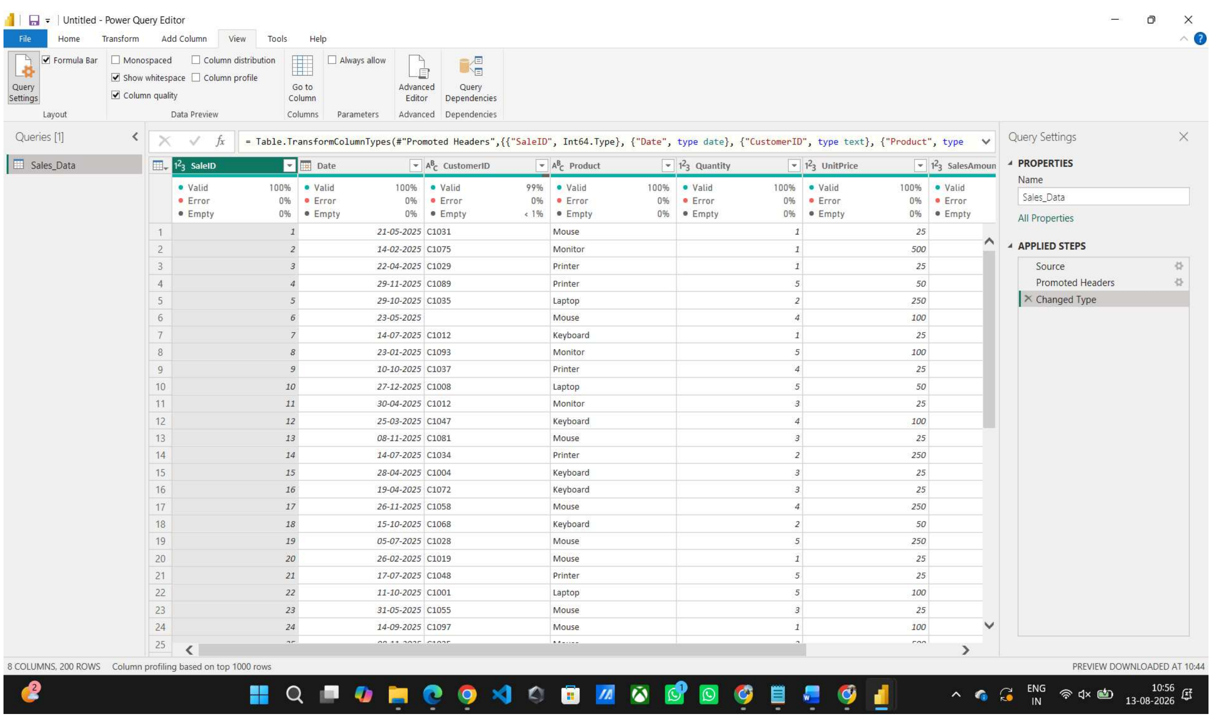


`Home > Get Data > Text/CSV`

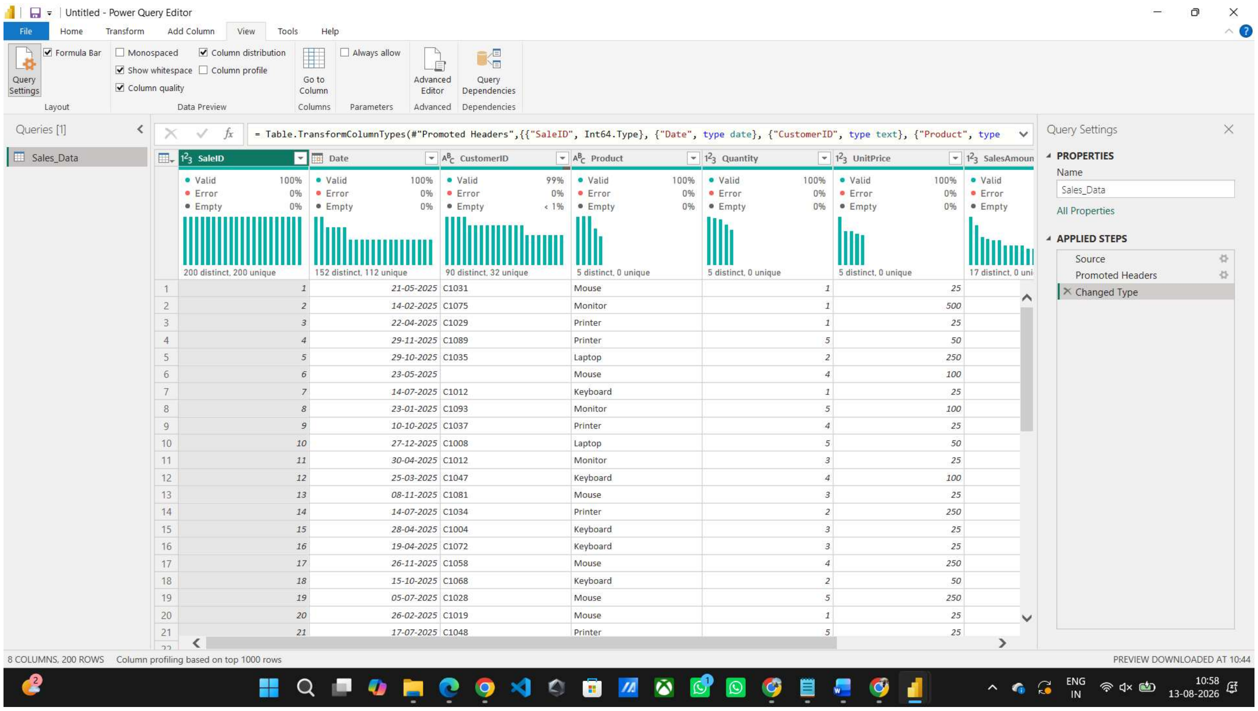


Browse to Sales_Data.csv > Load

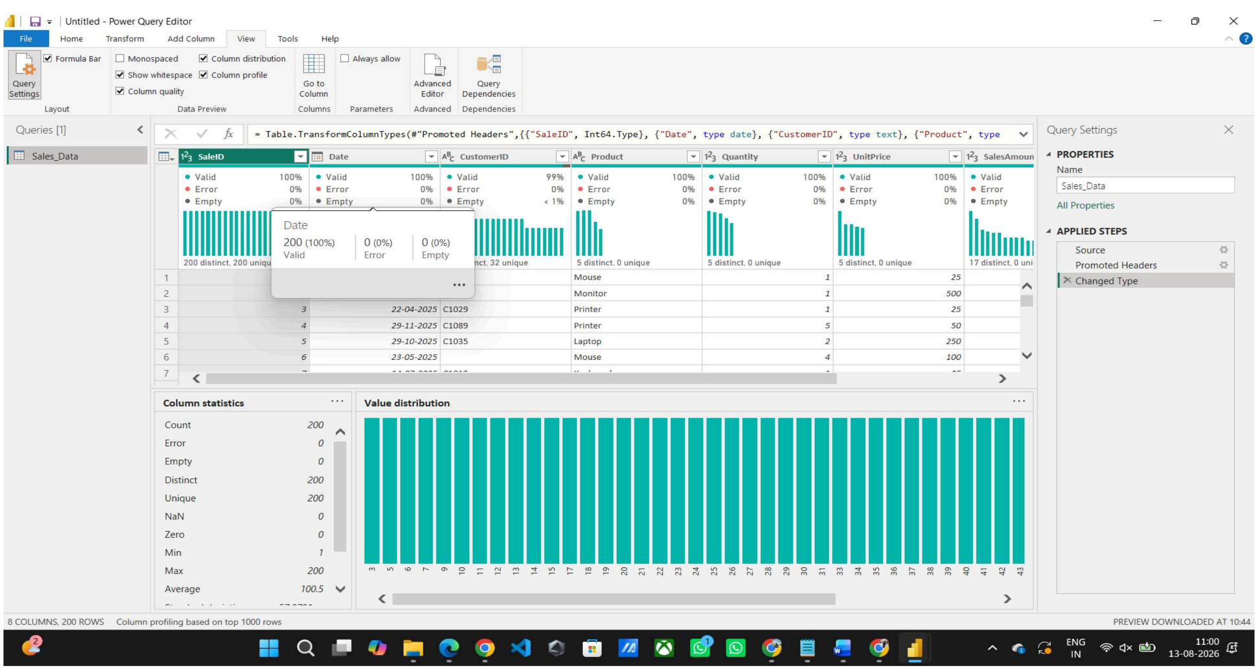
Step 4: Enable data profiling features.
View tab > Tick Column Quality



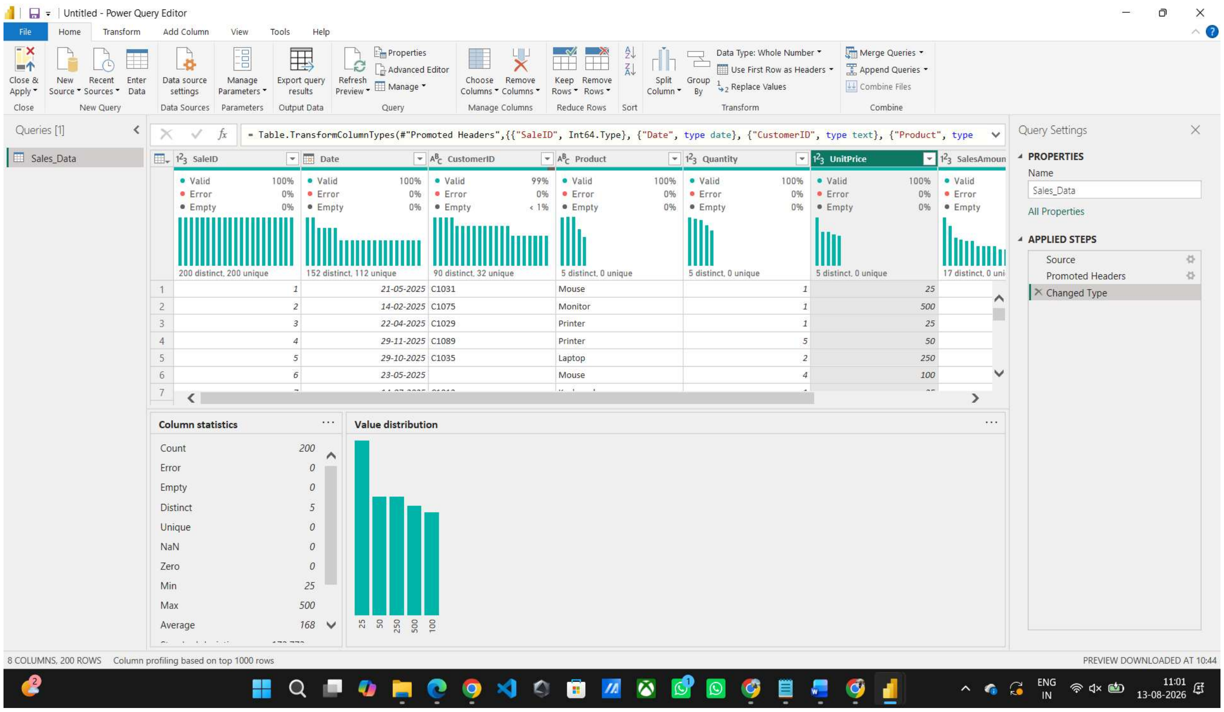
View tab > Tick Column Distribution



View tab > Tick Column Profile



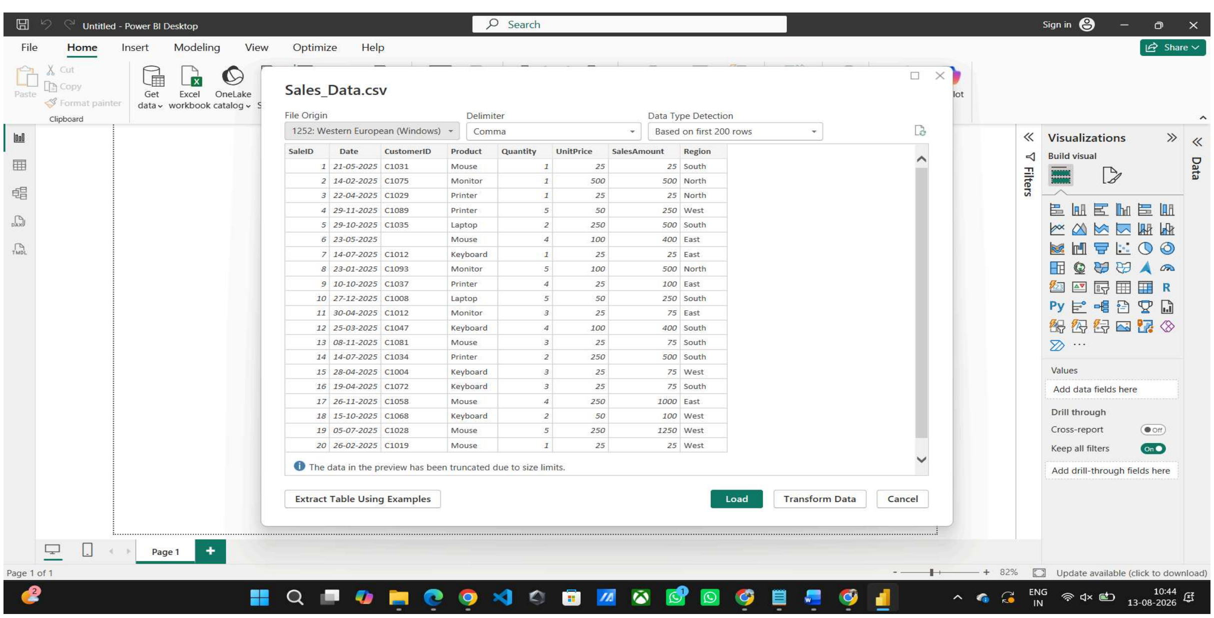
Select each column one at a time and observe the profiling statistics displayed below the data preview grid.



Part B: Data Profiling in Power Query Editor

Step 3: Open Power Query Editor.

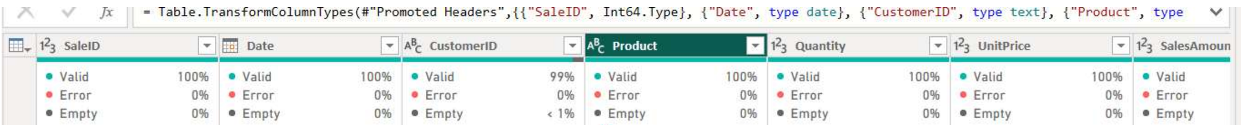
Home > Transform Data



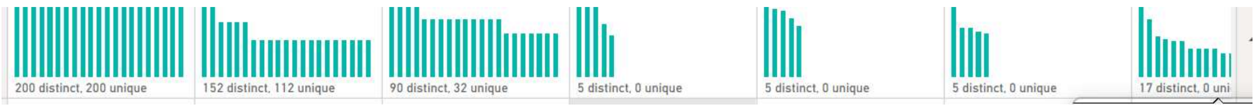
Step 5: Record profiling observations.

For each column in each dataset, observe and record:

- Data type and total row count.
- Null/empty percentage from Column Quality.



- Distinct and unique value counts from Column Distribution.



- Minimum, maximum, mean, and standard deviation for numeric columns from Column Profile.
- Any errors, mismatched data types, or unexpected values.

Part C: Identify and Document Data Quality Issues

Step 6: Identify data quality issues from the profiling results.

Check for common issues such as:

• High null or empty values in customer id columns.

• Columns containing only one unique value.

➔ There are no any columns containing one unique value.

• Numeric columns containing text values or incorrect data types.

• Date values outside the expected range or future dates.

• Duplicate records indicated by low uniqueness.

➔ Duplicate rows were identified in the dataset. The selected identifier column contains 200 rows, 90 distinct values, and 0 unique values, indicating substantial repetition. Duplicate records should be reviewed and removed or consolidated where appropriate.

Step 7: Document the identified issues. Record at least three data quality issues found during profiling and mention a suitable . 8.RESULT remediation action for each issue.

Result :

Data from multiple heterogeneous sources (CSV, Excel, and SQL database) was successfully acquired and imported into Microsoft Power BI Desktop. Data profiling was performed using the Power Query Editor's built-in tools — Column Quality, Column Distribution, and Column Profile. Key data quality dimensions including completeness, uniqueness, and validity were assessed across all datasets, and data quality issues were identified and documented for remediation prior to analysis.

Learning Outcomes:

- Explain the concept of Business Intelligence, its key components, and the role of data acquisition in a BI pipeline.
- Describe different data source types and the methods used to connect and import them into Power BI.
- Connect to multiple heterogeneous data sources and successfully import datasets using Power BI's Get Data feature.
- Apply data profiling techniques in Power Query Editor to assess data quality dimensions across real-world datasets.
- Interpret profiling statistics to identify data quality issues and propose suitable remediation strategies.